

Sampath Varma Datla

AI Engineer | LLM Systems & Agentic AI

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SUMMARY

AI Engineer with end-to-end ownership of LLM systems, RAG pipelines, and agentic AI (LangChain, LangGraph) in production on AWS. Spearheaded four AI and full-stack platforms across healthcare and ed-tech — cutting retrieval time by 70%, reducing false-positive flags by 45%, and scaling to 500+ concurrent users. Strong track record of translating complex technical solutions into measurable business impact

EDUCATION

GITAM University

Bachelor of Technology, Computer Science and Engineering

Sep. 2022 – Apr. 2026

EXPERIENCE

DATAi2i Private Limited

Junior AI Engineer | Promoted from AI Engineer Intern

May 2026 – Present

- Design and deploy LLM-powered NLP inference pipelines for pharmaceutical clients on AWS SageMaker and S3, reducing model deployment time by 30% and streamlining CI/CD workflows.
- Execute fine-tuning, evaluation, and optimization of large language models for domain-specific text classification, named-entity recognition, and contextual search, achieving a 25% accuracy gain over baseline.

ZenithZap Beverages Private Limited

AI Engineer & Full-Stack Developer Intern

Aug. 2025 – Oct. 2025

- Delivered a production e-commerce web application using React.js, Node.js, and Express.js, growing monthly active users past 5,000 within two months and achieving a 30% increase in brand visibility.
- Integrated a conversational AI chatbot via LLM APIs, boosting user engagement by 4%, decreasing support queries by 25%, and cutting response latency by 40% through server-side caching.

DATAi2i Private Limited

AI Engineer Intern

May 2025 – Jul. 2025

- Managed the full software development lifecycle of an internal knowledge assistant (Gemini, MongoDB, Streamlit), covering requirements analysis, system design, data pipeline setup, and production deployment.
- Accelerated employee document retrieval across 120+ files by 70% via semantic chunking, dense-vector indexing, and prompt calibration, achieving top-1 precision of 70% and removing 60% of off-topic results.

BMARG Innovative Solutions

Software Engineer Intern

Jan. 2025 – Apr. 2025

- Built a MERN-stack clinic management portal with Razorpay payment integration and Google Firebase Authentication, cutting patient onboarding time by 35% and improving page-load speed by 40%.
- Scaled the platform to 500+ concurrent sessions by profiling and resolving critical scheduling bottlenecks in collaboration with clinical staff.

PROJECTS

Enterprise Document Search Assistant | Python, Gemini, MongoDB, Streamlit, RAG, Vector Search

2025

- Constructed a RAG pipeline with dense-vector indexing and semantic segmentation, cutting staff document-search time by 70%, achieving top-1 retrieval precision of 70%, and removing 60% of off-topic results.

AI Fitness Coach | Python, LangChain, LangGraph, React.js, Node.js, OpenAI

2025

- Architected a multi-step agentic reasoning pipeline in LangGraph for personalized workout and nutrition plans, achieving 3× faster generation via structured goal-parsing and plan-synthesis stages.

LifePathBot — AI Student Success Platform | React, TypeScript, Vite, Tailwind CSS, Firebase, LLM APIs

2025

- Developed an AI-powered student success platform with an LLM-backed chat interface, multi-file ingestion (PDF, DOCX, PPTX), goal tracking, and an analytics dashboard; piloted with 200+ students.

Exam Integrity Monitor | JavaScript, WebRTC, MediaPipe, face-api.js, Computer Vision

2024

- Engineered a real-time 30 FPS webcam proctoring system using facial keypoint detection and anomaly scoring, reducing proctor workload by 60% across 1,000+ exam sessions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL

Web & Backend: React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs, HTML5, CSS3

AI & ML: Large Language Models (LLMs), LLM Fine-tuning, Retrieval-Augmented Generation (RAG), Vector Search, Embeddings, LangChain, LangGraph, TensorFlow, OpenCV, Natural Language Processing (NLP), Computer Vision, Prompt Engineering, MLOps, Machine Learning

Cloud & DevOps: AWS (SageMaker, S3), Docker, CI/CD, Firebase, MongoDB, MySQL, Supabase, Vercel

Tools: Git, GitHub, Postman, Streamlit, VS Code, Agile

ACHIEVEMENTS

- 1st Place** — Internal Smart India Hackathon (2025), GITAM University; outranked 400+ competing teams
- Selected** — IIT Delhi Computer Graphics & Computer Vision Summer School; chosen from 1,500+ applicants
- Top 10** — IIT Kanpur Techkriti Technical Innovation Event
- President** — GITAM Aero Astro Club (2025–26)
- Operations Lead** — IEEE CS GITAM Chapter (2024–25); organized technical workshops and coordinated cross-department initiatives